## C4.2 How do chemists explain the patterns in the properties of the elements?

1. describe the structure of an atom in terms of protons and neutrons in a very small central nucleus with electrons arranged in shells around the nucleus;
2. recall the relative masses and charges of protons, neutrons and electrons;
3. recall that in any atom the number of electrons equals the number of protons;
4. recall that all the atoms of the same element have the same number of protons;
5. recall that the elements in the modern Periodic Table are arranged in order of proton number;
6. recall that some elements emit distinctive flame colours when heated (for example lithium, sodium and potassium);
7. understand that the light emitted from an element gives a characteristic line spectrum;
8. understand that the study of spectra has helped chemists to discover new elements;
9. understand that the discovery of some elements depended on the development of new practical techniques (for example spectroscopy);
10. be able to use simple conventions (for example 2.8.1 or dots in circles) to represent the electron arrangements in the atoms of the first 20 elements in the Periodic Table;
11. recall that a shell (or energy level) fills across a period;
12. **understand that the chemical properties of an element are determined by its electron arrangement, illustrated by the electron configurations of the atoms of elements in groups 1 and 7.**